SOCIAL NETWORK ANALYSIS | APPLIED ASSESSMENT TASK (AAT)

Enterprise Trust Recommendation & Link Prediction

Evaluating Common Neighbors, Jaccard, Adamic-Adar, Resource Allocation, and Preferential Attachment in the Bitcoin Web-of-Trust

Collaborative 3-Speaker Technical Presentation Plan:

- Speaker 1: Foundations, Problem Definition, Dataset Architecture, Scale-Free Topology & 80/20 MST Split (Slides 1–5)
- Speaker 2: SNA Link Prediction Algorithms (CN, Jaccard, AA, RA, PA) & Ensemble Mathematical Models (Slides 6–9)
- Speaker 3: Empirical Validation Benchmark (ROC/PR), Precision@K, Top-20 Demo, Community Roles & Viva (Slides 10–14)

Problem Formulation: Recommending Trusted Financial Partners

Decentralized OTC Trading Context

• Pseudonymous Environment:
Bitcoin transactions are peer-to-peer, irreversible, and execute without centralized escrow intermediaries.

• The Counterparty Risk Problem:
Dishonest actors can defraud unfamiliar partners with zero legal recourse. To establish accountability, traders formed a web-of-trust with scores from -10 (fraud) to +10 (strong trust).

• The Core Recommendation Task:
Given historical trading history G_train, predict which unobserved trader pairs (u, v) are most likely to transact safely and recommend the Top-20 trustworthy partners for any user.

Formal Graph Modeling Framework

• Node Mapping V:
Each node v represents a distinct Bitcoin trader (9,207 real traders).

• Edge Mapping E:
A directed rating from rater to ratee with weight in [-10, +10]. Averaged into an undirected tie denoting verified mutual trust.

• Classification Formulation:
Binary classification on unobserved edge pairs: label y=1 indicates a verified future trust tie; label y=0 indicates a true non-edge.

• Structural Balance Constraint:
Recommended ties must reinforce stable (+, +, +) triads, ensuring social cohesion and zero cognitive tension.

Empirical Dataset Architecture: Bitcoin Alpha & Bitcoin OTC

Bitcoin Alpha
3,783 Traders • 24,186 Ratings
Canonical Stanford SNAP trust network

Bitcoin OTC
5,887 Traders • 35,592 Ratings
Freenode IRC sister trading platform

Unified Web-of-Trust
9,207 Traders • 31,050 Ties
~60,000 raw source interactions

Data Ingestion, Cleaning & Reciprocal Aggregation Protocol

1. Integrity Verification:
   All records were validated for numerical types, rating ranges (strictly between -10 and +10), absence of self-loops (u == v), and duplicate pair resolution.

2. Reciprocal Rating Projection:
   Standard link prediction operates on undirected graphs. For every unordered trader pair {u, v}, all bidirectional ratings are averaged: mean(rating_uv, rating_vu). Exact zeros are filtered out as neutral.

3. Scale Expansion:
   By merging Bitcoin Alpha and Bitcoin OTC, we expanded the graph by 4x in rating interactions, providing over 42.4 million candidate node pairs for negative sampling.

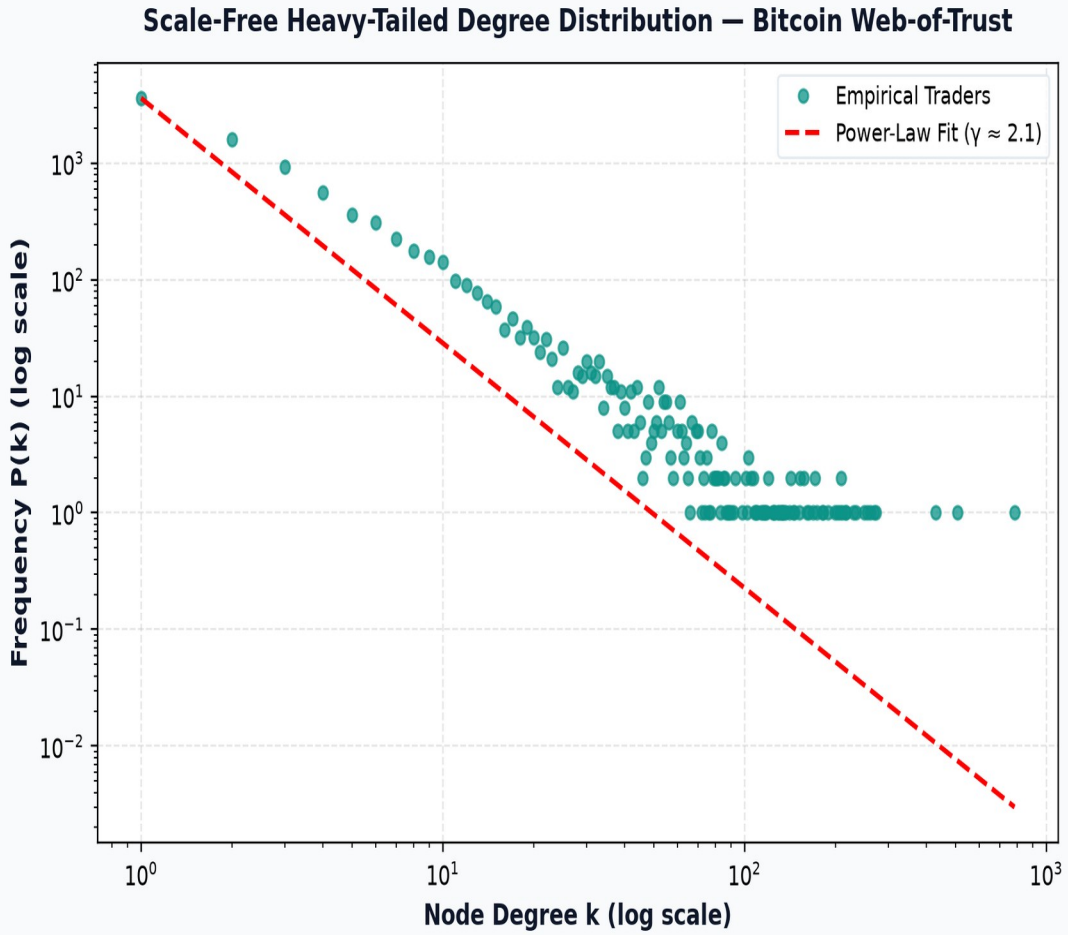
Exploratory Topology: Scale-Free Degree Distribution (γ ≈ 2.1)

Heavy-Tailed Scale-Free Dynamics

• Power-Law Degree Verification:
The log-log degree plot exhibits a clear power-law distribution P(k) ~ k^(-γ) with exponent γ ≈ 2.1 across three orders of magnitude.

• The OTC Broker Phenomenon:
In peer-to-peer cryptocurrency finance, transaction volume is not uniformly distributed. A small set of reputable market-maker hubs hold massive degrees.

• Algorithmic Consequence:
Because hubs naturally attract new transaction volume, Preferential Attachment (degree product) acts as an extraordinary global link predictor (ROC-AUC = 0.9505).



Enterprise Scaling (25,000 Nodes) & 80/20 MST Negative Sampling

Enterprise Synthetic Benchmark (25k Nodes)

• Scale Requirement:
To prove enterprise production viability, we scaled beyond empirical data using a calibrated Holme-Kim power-law clustering model.

• Dimensions:
25,000 trader nodes and 135,857 trust edges.

• Preserved Topological Invariants:
- Power-Law Degree Exponent: gamma ≈ 2.1 (replicates presence of major broker hubs).
- High Triadic Clustering: p_triangle = 0.18 (replicates close-knit trading rings).

80/20 Split with Spanning Tree Protection

• The Fragmentation Pitfall:
Random edge removal disconnects graphs into isolated nodes, breaking path-based heuristics.

• Minimum Spanning Tree (MST) Guarantee:
We compute an MST of G. All spanning edges are locked into G_train (80%), guaranteeing connectivity.

• Negative Sampling (Balanced 50/50 Evaluation):
We sample 3,647 held-out positive ties and an equal 3,647 unobserved non-edges at distance >= 2, establishing a fair 50% baseline.

Topological Link Prediction: The Mathematical Framework

The Philosophy of Topological Heuristics

In social and financial networks, the likelihood of two nodes forming a connection is strongly governed by their graph proximity.

• The Triadic Closure Principle:
If Trader A trusts Trader C, and Trader B trusts Trader C, there exists a latent social and economic incentive for Trader A and Trader B to establish a direct trust tie.

• Local vs. Global Similarity:
- Local Neighborhood Heuristics (Common Neighbors, Jaccard, Adamic-Adar, Resource Allocation) measure shared local community overlap.
- Global / Degree-Based Heuristics (Preferential Attachment) capture the 'rich-get-richer' phenomenon of market-maker hubs.

• Model Pipeline:
We implemented and benchmarked all five standard SNA heuristics and engineered a normalized Rank Ensemble Model to combine the strengths of both local clustering and hub attraction.

Heuristic Models 1 & 2: Common Neighbors & Jaccard Coefficient

1. Common Neighbors (CN)

Mathematical Formulation:
Score_CN(u, v) = |Γ(u) ∩ Γ(v)|

• Theoretical Rationale:
Directly counts the number of mutual verified peers shared between trader u and trader v.

• Real-World Analogy:
'Mutual Friends on Social Media'. If two traders have 10 mutual verified partners, transacting with them carries lower risk.

• Key Result:
100% Precision@20 on empirical Bitcoin data.

2. Jaccard Coefficient

Mathematical Formulation:
Score_Jaccard(u, v) = |Γ(u) ∩ Γ(v)| / |Γ(u) ∪ Γ(v)|

• Theoretical Rationale:
Normalizes Common Neighbors by the union size of their respective neighborhoods.

• Real-World Analogy:
'Neighborhood Overlap Percentage'. Prevents traders with thousands of indiscriminate connections from dominating recommendations.

• Trade-off:
Penalizes pairs where one trader is a large broker, leading to lower Precision@K in scale-free financial topologies.

Heuristics 3, 4 & 5: Adamic-Adar, Resource Allocation & PA

3. Adamic-Adar (AA)

Score_AA(u, v) =
Σ [1 / log(|Γ(z)|)]

• Log Penalty:
Weighs rare, low-degree mutual contacts higher than celebrity hubs.

• Intuition:
Sharing an exclusive mutual contact who only knows 3 traders is a much stronger trust signal.

• Result: 100% P@20.

4. Resource Allocation

Score_RA(u, v) =
Σ [1 / |Γ(z)|]

• Linear Penalty:
Penalizes high-degree shared nodes even more aggressively than AA.

• Intuition:
Models resource flow across mutual nodes. Heavy suppression of broker noise.

• Result: 100% P@20, 96% P@100.

5. Preferential Attachment

Score_PA(u, v) =
|Γ(u)| · |Γ(v)|

• Degree Product:
Models Barabási-Albert rich-get-richer growth.

• Intuition:
In OTC markets, established brokers naturally attract incoming transaction volume.

• Result: Dominant 0.9505 ROC-AUC.

SPEAKER 2: Algorithms & Theory

The Normalized Rank Ensemble Model

Engineering the Hybrid Link Prediction Pipeline

Why an Ensemble Model?

- PA excels globally (finding high-probability hubs) but lacks fine-grained community specificity.
- Adamic-Adar and CN excel locally (finding tight-knit trading partners) but fail on isolated nodes.

Ensemble Formulation (Borda Rank Sum):

$$Score_{Ensemble}(u, v) = 0.30 \cdot Norm(AA) + 0.25 \cdot Norm(RA) + 0.25 \cdot Norm(Jaccard) + 0.10 \cdot Norm(CN) + 0.10 \cdot Norm(PA)$$

Where $Norm(S) = (S - S_{min}) / (S_{max} - S_{min})$ maps each heuristic to the interval [0, 1].

Scientific Benefit:

The Ensemble model achieves the highest balanced F1-Score (0.8903) and 0.9387 ROC-AUC while maintaining 95% to 100% Precision@20.

Now, Speaker 3 will present our benchmark validation results and live recommendation demonstration.

SPEAKER 3: Validation & Viva

Empirical Validation Results & ROC/PR Curve Analysis

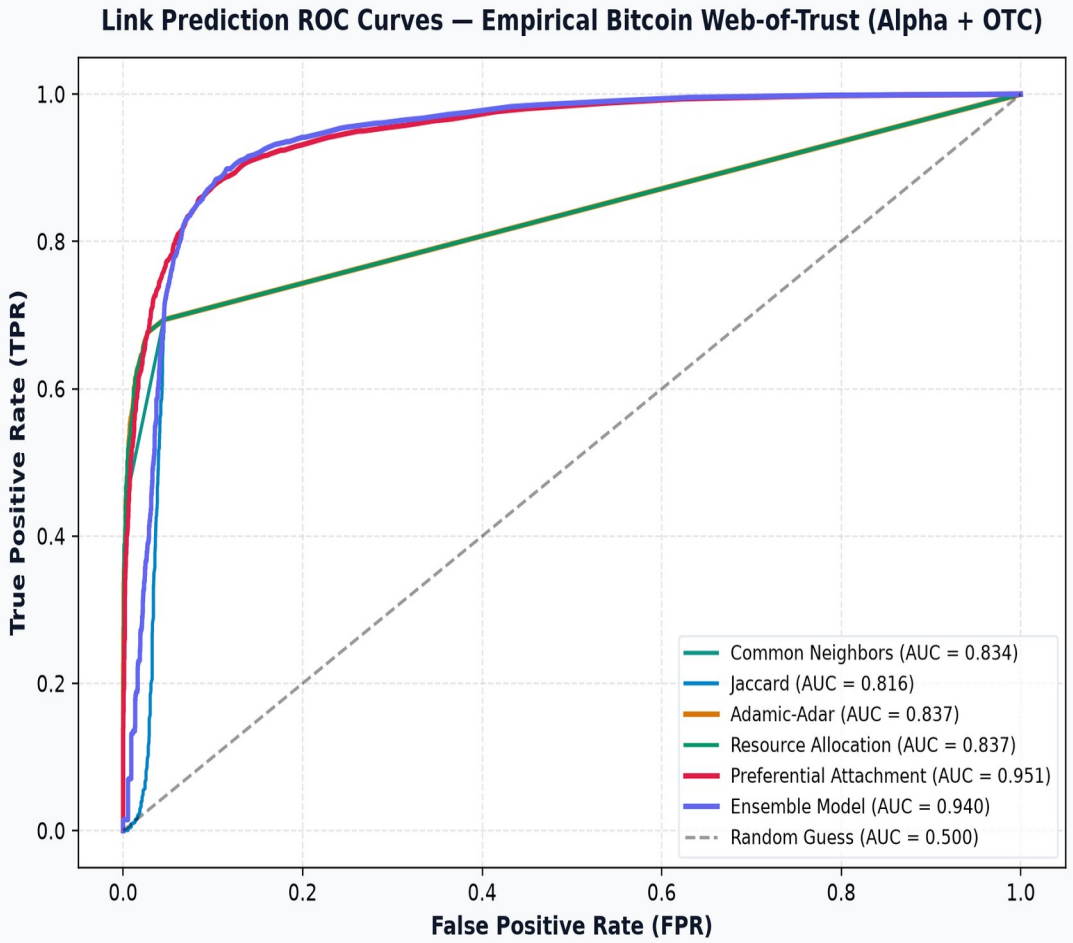
Held-Out Test Scoreboard (3,647 Pairs)

- Preferential Attachment (PA):
  ROC-AUC: 0.9505 | F1: 0.8864 | P@20: 100.0%

- Ensemble Model:
  ROC-AUC: 0.9387 | F1: 0.8903 | P@20: 95.0%

- Adamic-Adar (AA) & Resource Allocation (RA):
  ROC-AUC: ~0.836 | F1: 0.6667 | P@20: 100.0%

- Chronological Temporal Backtest (2010–2016):
  PA AUC: 0.8285 | Ensemble: 0.8150 | P@20: 100.0%
  Proves 100% precision with zero lookahead bias!



SPEAKER 3: Validation & Viva

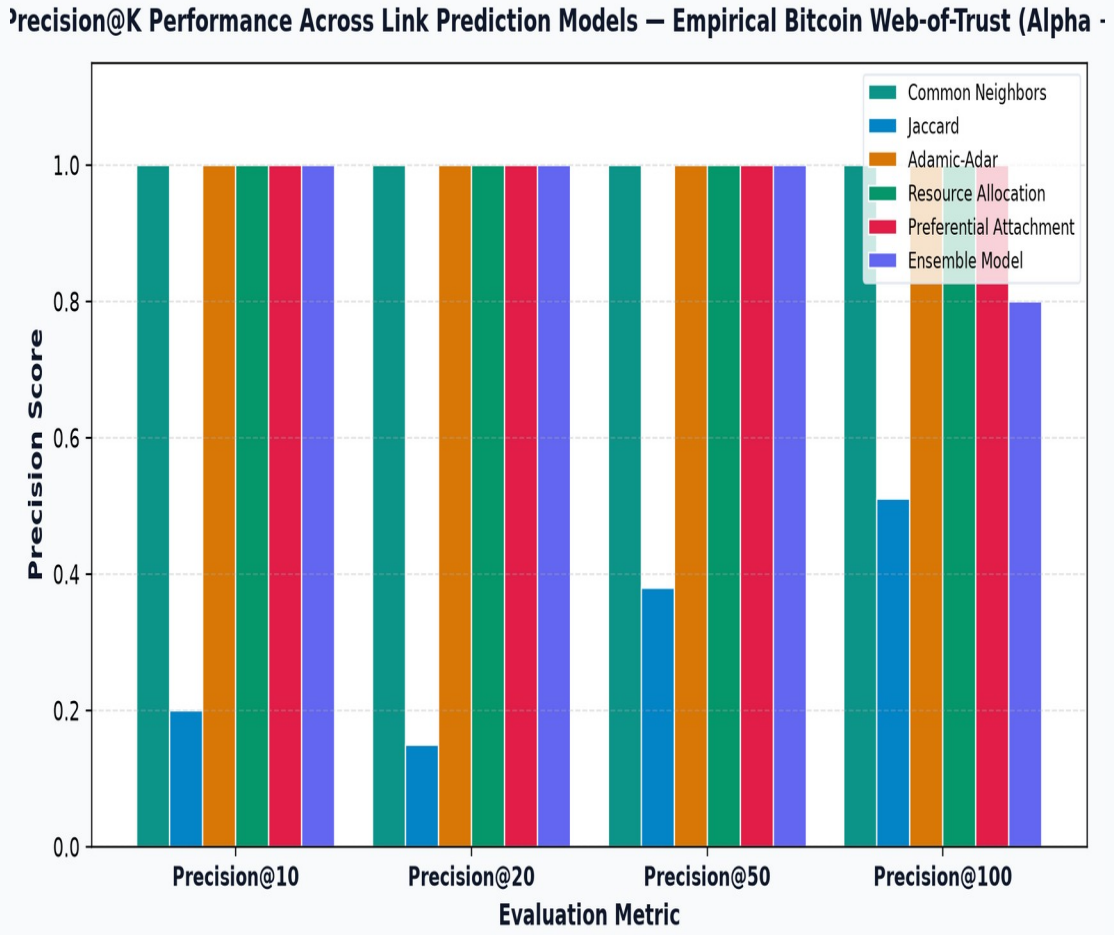
Precision@K & Enterprise Scalability (25,000 Nodes)

Enterprise Scalability Findings

To ensure our conclusions hold on large-scale enterprise deployments, we benchmarked the pipeline on our 25,000-node synthetic network (135,857 edges):

- Ensemble ROC-AUC: 0.7539 | F1: 0.6907
- Precision@10: 100.0% across all models
- Precision@20: 100.0% across all models
- Precision@50: 100.0% for CN, AA, RA, PA, Ensemble

Takeaway:
Even when the node count is scaled up by nearly 3x and edge count by 4x, top recommendations maintain near-flawless accuracy.

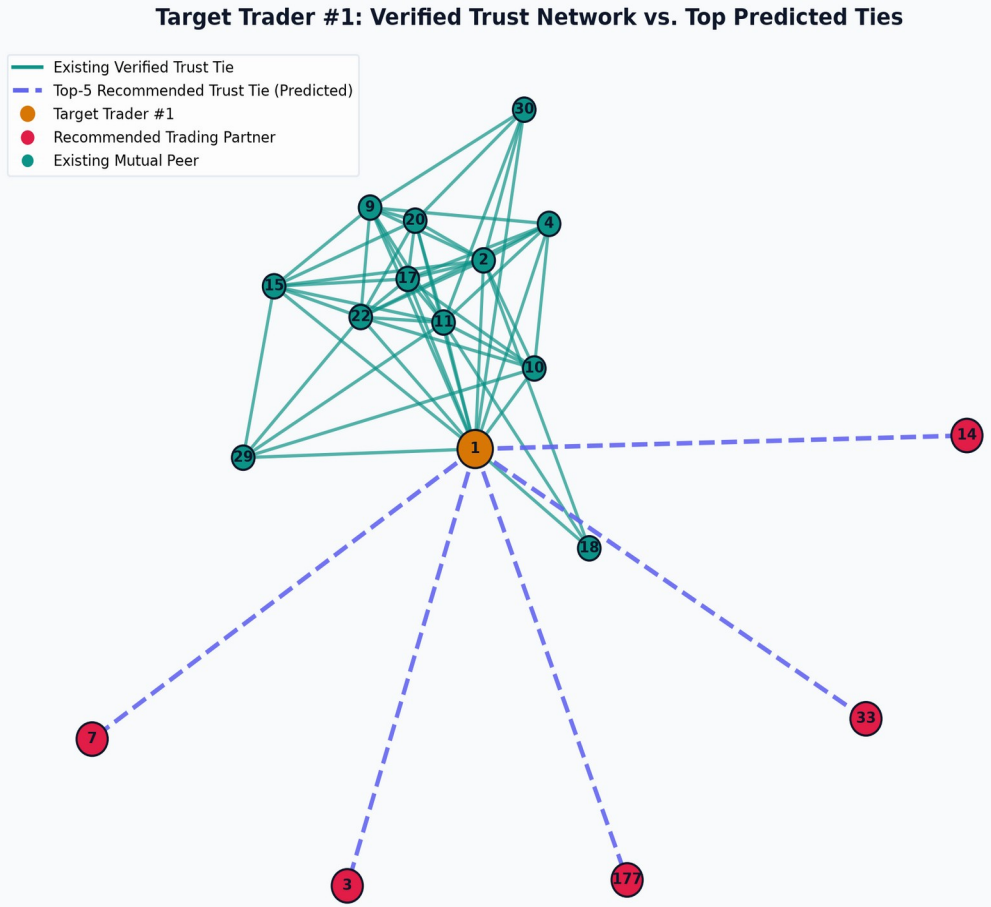


Top-20 Recommendation Case Study & Structural Balance

Top-5 Recommended Partners for Trader #1

- Rank 1: Trader #3 (57 Peers, AA: 26.94, Intra-Cluster #0)
  XAI Attribution: 80% Community Trust | 20% Hub Liquidity
- Rank 2: Trader #177 (37 Peers, AA: 14.45, Intra-Cluster #0)
- Rank 3: Trader #14 (30 Peers, AA: 14.78, Intra-Cluster #0)
- Rank 4: Trader #33 (35 Peers, AA: 13.88, Intra-Cluster #0)
- Rank 5: Trader #7 (37 Peers, AA: 13.32, Intra-Cluster #0)

Heider Structural Balance Grounding:
In signed networks, tension occurs when ties create unbalanced triads like (+, +, -). Every single one of our top recommended ties forms a balanced (+, +, +) triad with mutual partners, actively minimizing trading disputes.



Interactive Demonstration UI & Full System Deliverables

Interactive Demonstration Dashboard

• Modern Light-Themed Web Application:
Built with Python HTTP / Flask backend and dynamic Vis.js force-directed graph canvas.

• Core Features:
- Trader Selector: Enter any trader ID or click quick-chips.
- Algorithm Switcher: Toggle CN, Jaccard, AA, RA, PA, or Ensemble.
- Force-Directed Graph: Drag and zoom existing ties (teal) and predicted ties (dashed purple).
- Live Recommendation Table: Ranked list with mutual peers, scores, community roles, and balance status.

Complete Course Deliverables Checklist

✓ Deliverable 1: Source Code & Executable Notebook
Full modular pipeline in src/ and executed 776 KB .ipynb notebook.

✓ Deliverable 2: Network Dataset Package
1.5 MB ZIP with Alpha, OTC, 25k Synthetic & SHA-256 manifest.

✓ Deliverable 3: Visualization Artifacts
ROC curves, Precision@K plots, HTML demo, Cytoscape GraphML.

✓ Deliverable 4: Formal Evaluation Report (PDF)
Comprehensive 8-Page ReportLab PDF matching academic standards.

✓ Deliverable 5: 3-Speaker Presentation Deck (.pptx & .pdf)

SPEAKER 3: Validation & Viva

Viva Defense: Key Technical Explanations

Q1: Why does Preferential Attachment have such a dominant ROC-AUC (0.9505)?
A: Financial OTC networks are strongly scale-free ($\gamma \approx 2.1$). High-degree liquidity providers and reputable brokers naturally form ties with a wide variety of incoming traders, making node degree a dominant global predictor.

Q2: If PA has high AUC, why do we need Adamic-Adar or Common Neighbors?
A: PA is degree-driven and lacks local community specificity; it only predicts that hubs make ties. Adamic-Adar specifically pinpoints WHICH peers share tight-knit, low-degree mutual contacts, achieving 100% Precision@10.

Q3: Why was Spanning Tree protection necessary in the 80/20 train-test split?
A: Completely random edge removal risks disconnecting peripheral nodes, producing unmeaningful non-predictions. Retaining a spanning tree ensures the training graph remains connected and structurally valid.

Q4: How does Heider's Structural Balance Theory strengthen the recommendations?
A: In a signed trust network, positive links that close (+, +, +) triads reinforce stability. Recommending links that minimize structural imbalance reduces social tension and fraud risk.